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EDUCATION	Ph.D. in Computer Science	01/2024 – now	
	Johns Hopkins University		
	Affiliated with the Laboratory for Computational Sensing and Robotics.		
	Primary advisor: Mathias Unberath		
	M.S. Biomedical Engineering	08/2022 - 12/2023	
	Johns Hopkins University		
	B.S. Biomedical Engineering with Honors,	09/2018 – 06/2022	
	University of California, Davis		
SELECTED AWARDS	Selected Awards		
	10. Runner Up Best Paper Award	2026	
	For paper [C-6] at DT4H 2026.		
	9. Finalist, Best Paper Award	2026	
	For paper [J-5] at IPCAI 2026.		
	8. Finalist, Best Paper Award	2026	
	For paper [J-4] at IPCAI 2026.		
	7. Bench-to-Bedside Award	2024	
	For paper [J-1] at IPCAI 2024.		
	6. Finalist, Best Paper Award	2024	
	For paper [J-2] at IPCAI 2024.		
	5. LCSR Fellowship for Outstanding Incoming Ph.D. Students	2024	
	Johns Hopkins University		
	4. Best Project Award, Computer Integrated Surgery II	2023	
	Johns Hopkins University		
3. Best Project Award, Haptic Interface Design	2023		
Johns Hopkins University			
2. Excellence in Manufacturing Award	2022		
University of California, Davis			
1. Dean’s Honor List, University of California, Davis	2022		
Top 16% GPA in College of Engineering			

PUBLICATIONS I have (first/co)-authored 3/2 journal articles, 1/5 conference papers, and 2/1 preprints, and I am an inventor on 2 patents or patent applications in process. A detailed list is available below and on [Google Scholar](#).

Peer-reviewed Journal Articles

- [J-5]. **H. Zhang**, Y. Shen, R. D. Soberanis-Mukul, A. Ghosh, H. Ding, L. Seenivasan, J. L. Porras, Z. Mao, C. Li, W. Xiao, L. Yarmus, A. C. Argento, M. Ishii, M. Unberath. “TwinOR: Photorealistic Digital Twins of Dynamic Operating Rooms for Embodied AI Research” Accepted at *International Journal of Computer Assisted Radiology and Surgery*, 2026.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2026*
Audience vote for **long oral** presentation at IPCAI’26.
Finalist for **Best Paper Award** at IPCAI’26.
- [J-4]. B Inigo, S Kim, S.M. Cho, M. Song, **H. Zhang**, A. Uneri, C. Bailey, A. Krieger, M. Ishii, M. Unberath. “Vision-based Augmented Reality Guidance for Setting Up Robot-Assisted Spine Surgery” Accepted at *International Journal of Computer Assisted Radiology and Surgery*, 2026.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2026*
Finalist for **Best Paper Award** at IPCAI’26.
- [J-3]. **H. Zhang***, B.D. Killeen*, Y. C. Ku, L. Seenivasan, Y. Zhao, M. Liu, Y. Yang, S. Gu, A. Martin Gomez, R.H. Taylor, G. Osgood, M. Unberath. “StraightTrack: Towards Mixed Reality Navigation System for Percutaneous K-wire Insertion,” *Wiley Health Technology Letters*, 2024.
Special Issue: *Augmented Environments for Computer Assisted Interventions (AE-CAI) 2024*
- [J-2]. B.D. Killeen*, **H. Zhang***, L. Wang, Z. Liu, C. Kleinbeck, M. Rosen, R.H. Taylor, M. Unberath. “Stand in Surgeon’s Shoes: Virtual Reality Cross-training to Enhance Teamwork in Surgery,” *International Journal of Computer Assisted Radiology and Surgery*, 2024. (*Joint first authors)
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2024*
Audience vote for **long oral** presentation at IPCAI’24.
Finalist for **Best Paper Award** at IPCAI’24.
- [J-1]. C. Kleinbeck, **H. Zhang**, B.D. Killeen, D. Roth, M. Unberath. “Neural Digital Twins: Reconstructing Complex Medical Environments for Spatial Planning in Virtual Reality,” *International Journal of Computer Assisted Radiology and Surgery*, 2024.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2024*
Audience vote for **long oral** presentation at IPCAI’24.
Honored with the **Bench-to-Bedside Award** at IPCAI’24.

Peer-reviewed Conference Papers

- [C-6]. **H. Zhang***, B. Yang*, M. Unberath. “Patient-Specific Articulated Digital Twins from a Single Full-Body CT Scan.” *International Workshop on Digital Twin for Healthcare (DT4H), MICCAI 2026 Workshop*.
Runner Up Best Paper Award at DT4H’26.
- [C-5]. Y. Shen, **H. Zhang**, M. Unberath. “Training LLMs with Reinforcement Learning over Digital Twin Representations for Reasoning-Intensive Surgical VideoQA”. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.
- [C-4]. H. Ding, L. Seenivasan, H. Shu, G. Byrd, **H. Zhang**, P. Xiao, J. Barragan, R. H. Taylor, P. Kazanzides, M. Unberath. “Towards Robust Automation of Surgical Systems via Digital Twin-based Scene Representations from Foundation Models”. *COLlaborative Intelligence and Autonomy in Image-guided Surgery (COLAS), MICCAI 2025 Workshop*
- [C-3]. B.D. Killeen, L.J. Wang, B. Inigo, **H. Zhang**, M. Armand, R.H. Taylor, G. Osgood, M. Unberath. “FluoroSAM: A Language-promptable Foundation Model for Flexible X-ray Image Segmentation.” *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
- [C-2]. A. Perez, **H. Zhang**, Y.C. Ku, L. Seenivasan, R. D. Soberanis-Mukul, J. L. Porras, R. Day, J. Jopling, P. Najjar, M. Unberath. “Privacy-Preserving Operating Room Workflow Analysis using Digital Twins,” *Medical Imaging with Deep Learning (MIDL)*, 2025.
- [C-1]. B.D. Killeen, **H. Zhang**, J.E. Mangulabnan, M. Armand, R. Taylor, G. Osgood, M. Unberath. “Pelphix: Surgical Phase Recognition from X-ray Images in Percutaneous Pelvis Fixation,” *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023.
[Featured in the JHU Hub and Surgery International.](#)

Preprints

- [M-3]. Nelson, N., Chen, J. T., Haworth, J., Chen, X., Zbinden, L., Huang, D., ..., Speidel, S.. “Open-H-Embodiment: A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics.” *arXiv*, 2026, 2604.21017.
- [M-2]. **H. Zhang***, S. M. Cho*, J. E. Mangulabnan*, Z. Mao, Y. He, P. Guo, D. Xu, G. Hager, M. Ishii, M. Unberath. “Humanoid Robots as First Assistants in Endoscopic Surgery.” *arXiv*, 2026, arXiv:2602.24156. (*Joint first authors)
- [M-1]. **H. Zhang**, L. Seenivasan, J. L. Porras, R. D. Soberanis-Mukul, H. Ding, H. Shu, B.D. Killeen, A. Ghosh, L. Yarmus, M. Ishii, A. C. Argento, M. Unberath. “Did You Just See That? Arbitrary View Synthesis for Egocentric Replay of Operating Room Workflows from Ambient Sensors.” *arXiv*, 2025, arXiv:2510.04802.

TEACHING	Computer Integrated Surgery II EN.601.456/656 , Project Mentor Johns Hopkins University	
	– <i>Measuring Variability of Pelvic Standard Views in Virtual Reality</i>	2024
	Voted runner-up, Best Project Award.	
	– <i>A Cannula Marker Body for Tracker-free Surgical Navigation during Kirschner Wire Placement</i>	2024
	Computer Integrated Surgery I EN.601.455/655 , Course Assistant Department of Computer Science, Johns Hopkins University	Fall 2024
	Haptic Interface Design EN 530.491/691 , Teaching Assistant Department of Mechanical Engineering, Johns Hopkins University	Fall 2023
	Introduction to Augmented Reality EN 601.454/654 , Course Assistant Department of Computer Science, Johns Hopkins University	Fall 2023
TALKS AND PRESS	Invited Talks and Posters	
	[P-5]. Physics and AI-Driven Digital Twins, Selected Poster Hamlyn Symposium on Medical Robotics, UK <i>“Digital Twin Virtual Reality Training for Central Line Dressing Change”</i>	06/2026
	[P-4]. End of Semester Social, Selected Posters and Demos Data Science and AI Institute, Johns Hopkins University, USA <i>“Interventional X-ray Imaging in Virtual Reality for Orthopedic Surgery”</i>	05/2024
	[P-3]. IEEE World Haptics Conference Delft, Netherlands <i>“3D Hapkit: 3-degree-of-freedom (DOF) Haptic Device using a Delta Parallel Mechanism”</i>	07/2023
	[P-2]. LCSR Industry Day Johns Hopkins University, USA <i>“PelvisVR: Recreating Pelvic Trauma Surgery in Virtual Reality”</i>	04/2023
	[P-1]. College of Engineering Design Showcase University of California, Davis, USA <i>“THF:Radiolucent Hand and Wrist Fixation Device for Intraoperative Fluoroscopy”</i>	06/2022
	Selected Press	
	[P-3]. Our humanoid surgical-assistant robot work [M-2] was featured in <i>The Baltimore Banner</i> .	
	[P-2]. Our work [J-2] and [J-1] were featured in a JHU Computer Science News article on the IPCAI 2024 conference.	

[P-1]. Our work [C-1] presenting the first approach to surgical phase recognition in X-ray guided surgery with dynamic simulation was featured in the [JHU Hub](#) and [Surgery International](#).

SERVICE

Conference Reviewer

- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- International Conference on Information Processing in Computer-Assisted Interventions (IPCAI)
- International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)